scaling efficiency, we want to reduce the amount of communication between nodes as much as possible, both in terms of the number of times nodes need to send data as well as the amount of data that they send.

6.5.4. Communicating between nodes

The researchers at OpenAI developed a neat strategy for distributed computing where each node sends only one number (not a whole vector) to each other node, eliminating the need for a separate master node. The idea is that each worker is first initialized with the same parent parameter vector. Then each worker adds a noise vector to its parent to create a slightly different child vector (figure 6.15). Each worker then runs the child vector through the environment to get its fitness score. The fitness score from each worker is sent to all other workers, which just involves sending a single number. Since each worker has the same set of random seeds, each worker can recreate the noise vectors used by all the other workers. Lastly, each worker creates the same new parent vector and the process repeats.

Figure 6.15. The architecture derived from OpenAI's distributed ES paper. Each worker creates a child parameter vector from a parent by adding noise to the parent. Then it evaluates the child's fitness and sends the fitness score to all other agents. Using shared random seeds, each agent can reconstruct the noise vectors used to create the other vectors from the other workers without each having to send an entire vector. Lastly, new parent vectors are created by performing a weighted sum of the child vectors, weighted according to their fitness scores.